

4	(a)	In a polarised wave, the <u>vibrations of wave particles are limited to only one axis</u> ; whereas an unpolarised wave is not (i.e. no specific axis of vibrations or many different axis of vibrations).	B1
	(b)(i)	0 as no light will be able to pass through the polariser given it is perpendicular.	A1
	(b)(ii)1.	$A_3 = A_2 \cos(62^\circ - 23^\circ)$ $A_2 = A_1 \cos(23^\circ)$ thus $A_3 = A_1 \cos(23^\circ) \cos(62^\circ - 23^\circ)$ $= 0.715 A_1$	M1 M1 A0
	(b)(ii)2.	let the initial amplitude (unpolarised light) = A_0 $\frac{1}{2} = \left(\frac{A_1}{A_0}\right)^2$ $A_0 = \sqrt{2} A_1 = 1.41 A_1$ $\frac{I_3}{I_0} = \left(\frac{A_3}{A_0}\right)^2 = \left(\frac{0.715 A_1}{\sqrt{2} A_1}\right)^2 = 0.256$ Percentage of intensity reduced = $(1 - 0.256) \times 100\% = 74.4\%$	C1 C1 A1

5	(a)(i)		A1
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